EXPERIMENT 6

DATA VISUALIZATION USING POWER BI

Aim

To import, transform, analyze, and visualize the Employee Salary dataset using Power BI Desktop and create interactive dashboards and reports for salary analysis and decision-making.

Algorithm 1

Importing Data into Power BI

Step 1 Open Power BI Desktop.

Step 2 Select: Home → Get Data.

Step 3 Choose Text/CSV or the format of the Employee Salary dataset.

Step 4 Browse and select the employee salary dataset.

Step 5 Load the dataset and verify the imported fields.

Dataset: Employee Salary Analysis Dataset

Algorithm 2

Data Transformation using Power Query

Step 1 Select Transform Data.

Step 2 Open Power Query Editor.

Step 3 Remove duplicate employee records where applicable.

Step 4 Handle missing/null values according to the analysis requirement.

Step 5 Rename or standardize columns where required.

Step 6 Set appropriate data types for salary, age, experience, department and other fields.

Step 7 Apply changes.

Procedure

1. The data preview window will appear. Select Transform Data instead of Load to open the Power Query Editor.

2. Examine the imported table and verify important fields such as Employee ID, Employee Name, Department, Job Role, Education, Experience, Age, Location and Salary.

3. Remove completely blank rows using Home → Remove Rows → Remove Blank Rows.

4. Select the required columns and choose Home → Remove Rows → Remove Duplicates.

5. Standardize text fields such as Department, Job Role and Location so that values use a consistent representation.

6. Verify that salary and experience-related fields have suitable numeric data types.

7. Handle missing numeric values according to the analysis requirement. For salary visuals, exclude blank salary measurements rather than assigning an artificial salary.

8. Review the Applied Steps pane and correct any transformation errors before applying the query.

9. After completing cleaning and transformation, select Home → Close & Apply.

10. Switch to Data/Table view and inspect the transformed employee salary data.

Recommended data types

Column	Data Type
Employee ID	Text / Whole Number
Employee Name	Text
Department	Text
Job Role	Text
Education	Text
Experience	Whole Number / Decimal Number
Age	Whole Number
Location	Text
Salary	Decimal Number

Algorithm 3

Creating a Department-wise Salary Bar Chart

Step 1 Choose Bar Chart or Clustered Column Chart visualization.

Step 2 Drag Department → X-Axis / Y-Axis.

Step 3 Drag Salary → Values and set aggregation to Average.

Step 4 Format the chart and sort by Average Salary.

Interpretation: The department-wise chart helps identify departments with higher or lower average salaries.

Algorithm 4

Creating a Salary Distribution Chart

Step 1 Select a suitable column/bar chart.

Step 2 Create salary bands or use the Salary field to group employees into salary ranges.

Step 3 Drag the salary range to the Axis.

Step 4 Drag Employee ID to Values and use Count.

Step 5 Format data labels where required.

Interpretation: The salary distribution shows how employees are distributed across different salary ranges.

Additional Employee Salary Analysis

Create charts to compare average salary by Job Role, Education, Experience level and Location. A line chart can also be created using Experience on the X-axis and Average Salary on the Y-axis to observe the relationship between experience and salary.

Algorithm 5

Creating KPI Cards

Create measures using New Measure from the Measure tools tab.

Measure	DAX Formula
Total Employees	COUNTROWS('EmployeeSalary')
Average Salary	AVERAGE('EmployeeSalary'[Salary])
Maximum Salary	MAX('EmployeeSalary'[Salary])
Minimum Salary	MIN('EmployeeSalary'[Salary])
Average Experience	AVERAGE('EmployeeSalary'[Experience])
Department Count	DISTINCTCOUNT('EmployeeSalary'[Department])

Step 1 Create Card visuals for each measure.

Step 2 Drag the corresponding measure into the field well.

Output:

The KPI cards summarize the number of employees and key salary statistics such as average, maximum and minimum salary. These KPIs provide a quick overview of the employee salary dataset.

Algorithm 6

Creating Slicers

Step 1 Select Slicer Visualization.

Step 2 Drag Department into the slicer.

Step 3 Enable Single Selection when a single-department analysis is required.

Step 4 Apply filters.

Selecting a department in the slicer filters the KPI cards and charts on the report page.

Algorithm 7

Creating Dashboard

Step 1 Add KPI Cards for Total Employees, Average Salary, Maximum Salary and Minimum Salary.

Step 2 Add Department-wise Average Salary Bar Chart.

Step 3 Add Salary Distribution Chart.

Step 4 Add Job Role / Experience Salary Chart.

Step 5 Add Department or Location Slicer.

Step 6 Arrange and format visuals.

Step 7 Save the Power BI report/dashboard.

Result

The Employee Salary data was successfully prepared for import, transformation, analysis, and visualization in Power BI Desktop. Interactive KPI cards, department-wise salary comparison, salary distribution, experience-based analysis, and slicers were created to support effective employee salary analysis and decision-making.